Xu Han

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Photonics M.Sc. student and LMU astrophysics graduate working on wavefront sensing, optical modelling and experimental instrumentation.

Education

M.Sc. Photonics *Mar 2026 - Present*
Hochschule München (Munich University of Applied Sciences)
Remaining requirements are the project study and master's thesis. Expected completion in 2027. **Selected coursework:** Physical Optics and Modelling (**Zemax optical design**); Micro and Fiber Optics; Optical Measurement Technology; Laser Physics — Non-linear Optics — Ultrafast Optics; Semiconductor Optics and Components.

M.Sc. Astrophysics | LMU Munich *Apr 2022 - Aug 2024*
Grade: 1.55 (German scale). **Selected coursework:** Essentials of Advanced Astrophysics (1.3); Cosmology and Structure Formation (1.0); Statistical Methods; Instrumental Laboratory Courses. Degree certificate

B.Sc. Physics | University of Duisburg-Essen *Oct 2018 - Mar 2022*
Grade: 1.8 (German scale). Degree certificate

B.Eng. Electrical Engineering and Automation *Sep 2013 - Jul 2017*
Hubei University of Technology
Electronics, control systems and embedded systems.

Selected Research and Engineering Projects

Shack-Hartmann Wavefront Sensing & Adaptive Optics Simulation

Independent project | Interactive explorer | Source code *Mar 2026 - Present*

- Developed a modular Python SCAO framework connecting atmospheric turbulence, detector-level Shack-Hartmann sensing, calibrated reconstruction, deformable-mirror correction and delayed feedback.
- Traced four faint spots in a simulated 42 m telescope image to waffle modes the sensor barely detects. Filtering those modes reduced their mean brightness from 10.8 to 0.91 times the local halo level.
- Diagnosed a reconstruction failure associated with measurement dropout in a 2.2 m NGS model; stronger TSVD regularisation kept reconstruction usable throughout the matched two-second run.
- Cross-checked selected optical components against HCIPy and analytic references; documented reproducible experiments, model limitations and an interactive explorer of recorded results.

Optical Reconstruction of ESO's 100 m OWL Telescope *2026*

Independent extension of a Zemax laboratory project

- Rebuilt the proposed OverWhelmingly Large Telescope from ESO's Blue Book prescription. At 5 arcmin off axis, the RMS wavefront error was 1.472 µm, within 0.3% of ESO's 1.476 µm.
- At 3 arcmin, traced a 27-percentage-point discrepancy in ensquared energy to the focal surface. A spherical surface brought the model within 0.4 percentage points of ESO's sampled curve.

Weak Lensing Void Finder *Mar 2023 - Aug 2024*

Master's thesis | LMU Munich Supervisors: Dr. Chris Davies and Prof. Joseph Mohr.

- Developed a watershed void finder for 2D full-sky weak-lensing maps and tested its parameter sensitivity. Measured void abundance, effective radii, ellipticities and convergence profiles in DES Y3 data.
- Trained Gaussian-process emulators (scikit-learn) on simulations of 2,500 cosmologies and used them with MCMC for parameter inference.

Experimental Collisional Charging Apparatus *Mar 2021 - Aug 2021*

Bachelor's thesis | University of Duisburg-Essen Supervisors: Prof. Gerhard Wurm and Dr. Felix Jungmann.

- Designed the mechanical apparatus in CAD, fabricated components by 3D printing and assembled the setup; integrated and debugged electronic charge-measurement modules.
- Conducted measurements and analysed grain-size dependence of collisional charging, testing a dust-collision growth hypothesis relevant to early planet formation.

Laboratory and Instrumentation Training

- **Astronomical instrumentation laboratory:** Closed the Shack–Hartmann loop on LMU's ALPAO bench in July 2022, first correcting static aberrations with the deformable mirror, then correcting turbulence introduced by a screen. Other labs covered CCD imaging and reduction, optical interferometry and radio measurements of Galactic rotation.
- **Photonics laboratory:** Measured laser beam quality (M^2), shaped ultrashort pulses and characterised them by autocorrelation. Fibre experiments covered OTDR, polarisation, Bragg gratings and coupling. Also worked on APD measurements, laser spectra, laser interferometry and nonlinear optics.
- **Observatory experience:** Took and processed images with my supervisor at Wendelstein (43 cm).

Additional Independent Projects

Astrophotonics

- Modelled how AO wavefront error, injection jitter and photonic-lantern mode count affect fibre coupling, spectral SNR and radial-velocity precision.

Work Experience and Career Break

Python Developer | BaryonBib *Dec 2024 - Jan 2025*

MAX!mize start-up team, MPA, Garching

- Implemented and tested LLM features for the team's centralised Python bibliography platform.

Medical leave *Jan 2025 - Dec 2025*

Recovery and rehabilitation following an accident and surgery.

Talks, Workshops and Schools

MAVIS 2026 workshop | Participant *Sep 2026*

Tonale Winter School on Cosmology | Participant *Dec 2024*

ORIGINS Lensing Day *Nov 2024*

Contributed talk: Watershed on the curved sky

Cambridge-LMU Cosmology Meeting | Participant *Dec 2023*

Advanced Astrophysical Seminar | LMU Munich *Nov 2022*

Exploring Telescope Optics: How Do the Largest Optical Systems Work? A 45-minute presentation followed by 15 minutes of questions. Supervisor: Dr. Frank Grupp.

Technical Skills

Optical modelling: Wavefront sensing, Fourier PSFs, calibration, regularised reconstruction and closed-loop diagnostics; HCIPy and TIPTOP.

Zemax OpticStudio: telescope prescription reconstruction, PSF and ensquared-energy analysis

Scientific computing: Python (NumPy, SciPy, Matplotlib, pandas, scikit-learn); MATLAB; Mathematica; statistical analysis and MCMC.

Experimental methods: CAD, 3D printing, mechanical assembly, electronic measurement integration, optical alignment and interferometry.

Research tools: Linux, Git, Jupyter and LaTeX.

Languages and Interests

Languages: Chinese (native); English (fluent); German (fluent); French (beginner).
Interests: Astronomical photography and alpine mountaineering.

Academic References

Dr. Chris Davies — direct supervisor for my M.Sc. thesis.
Prof. Joseph Mohr — M.Sc. thesis supervisor. Taught Cosmology and Structure Formation and co-taught Essentials of Advanced Astrophysics.